SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)*

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Case study

Code of Conduct:

I _Vallapu Mohan Kumar (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understa ndi ng of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Applicat ion of Concept s	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts
A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.
Questions:

A. How can region-based representation and medial axis computation help in identifying defective parts?
Answer:
Region-Based Representation

- Region-based representation describes an object using all the pixels inside its boundary.
- It measures properties such as **area, perimeter, centroid, orientation, compactness, and eccentricity**.
- A normal gear has a fixed shape and region properties.
- If a gear has a missing tooth or extra protrusion, these region properties change.
- By comparing the measured properties with those of a standard component, defective parts can be identified automatically.

Medial Axis Computation (Skeletonization)

- The medial axis is the centerline (skeleton) of a shape while preserving its topology.
- It reduces the object into a thin representation without changing its structure.
- It helps detect:
  - Missing gear teeth
  - Broken edges
  - Extra protrusions
  - Shape deformation
- Comparing the skeleton of a test component with the skeleton of a normal component makes defect detection easier.

Conclusion:
Region-based representation identifies changes in object properties, while medial axis computation highlights structural defects, making defect detection more accurate.

B. Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.
Answer:
Effect of Noise
Noise introduced during image acquisition can affect shape analysis by:

- Producing false boundaries.
- Creating unwanted protrusions.
- Hiding small defects.
- Distorting object edges.
- Causing incorrect segmentation.
- Reducing measurement accuracy.

Preprocessing Steps

1. Noise Filtering

- Apply Gaussian Filter to remove Gaussian noise.
- Apply Median Filter to remove salt-and-pepper noise.

2. Thresholding

- Convert grayscale images into clean binary images.

3. Morphological Operations

- Erosion removes small unwanted pixels.
- Dilation restores object size.
- Opening removes tiny objects.
- Closing fills small holes.

4. Edge Detection

- Use Canny edge detection to obtain clear object boundaries.

5. Region Filling

- Fill holes inside the object to produce complete regions.

6. Image Normalization

- Adjust brightness and contrast for consistent analysis.

Conclusion:
Proper preprocessing removes noise and improves the accuracy of contour extraction, region analysis, and defect detection.

C. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.
Answer:

Fourier Descriptors	Wavelet Descriptors
Represent shape using frequency components.	Represent shape using multiple resolutions.
Good for global shape analysis.	Good for both global and local shape analysis.
Rotation and scale invariant.	Handles scale variation effectively.
Less effective for detecting small local defects.	Excellent for identifying tiny cracks and missing parts.
Faster computation.	Slightly higher computational cost.
Suitable for smooth shapes.	Suitable for irregular and complex shapes.

Conclusion

- Fourier descriptors work well for overall shape comparison.
- Wavelet descriptors detect local defects more accurately.
- For industrial defect detection, **wavelet descriptors provide better classification performance** because they capture both global and local shape information.

Case Study 2: Medical Image Segmentation
A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients. **Questions:**

A. Explain how active contours (snakes) can be applied to segment the organ boundaries.
Answer:
Active Contours (Snakes) are deformable curves used to locate object boundaries.
Working Process

1. Place an initial contour near the organ.
2. Internal forces keep the contour smooth.
3. External image forces attract the contour toward image edges.
4. The contour gradually moves until it reaches the organ boundary.
5. The final contour represents the segmented organ.

Advantages

- Produces accurate organ boundaries.
- Handles irregular shapes.
- Gives smooth segmentation.

- Works well for CT and MRI images.
- Can adapt to moderate shape variations.

Conclusion:
Active contours accurately segment organs by continuously deforming until they fit the object boundaries.

B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.
Answer:
Challenges:
1. Poor Initialization
- If the initial contour is far from the organ, it may converge to the wrong boundary.
- Incorrect initialization increases segmentation errors.

2. Weak Edges
- CT images may have blurred or low-contrast boundaries.
- Snakes may stop before reaching the actual edge.

3. Image Noise
- Noise can attract the contour to false edges.

4. Parameter Selection
Proper values must be chosen for:
- Elasticity
- Rigidity
- External force weight
- Step size

Improper values may cause:
- Over-smoothing
- Boundary leakage
- Slow convergence
- Incorrect segmentation

Solutions
- Apply image smoothing before segmentation.
- Use automatic contour initialization.
- Adjust parameters experimentally.
- Employ multi-scale segmentation.

Conclusion:
Careful contour initialization and parameter tuning significantly improve segmentation accuracy.

C. How could level set methods improve segmentation for cases where organs split or merge in the images?

Answer:

Level set methods represent contours using a mathematical function instead of explicit curves.

Advantages

- Automatically handles splitting of contours.
- Automatically handles merging of contours.
- Segments organs with complex topology.
- Provides smooth and accurate boundaries.
- More robust against image noise.
- Can detect multiple organs simultaneously.
- Easily adapts to large shape variations.

Example:

During liver segmentation:

- Blood vessels may divide the liver into multiple regions.
- Active contours struggle to split naturally.
- Level set methods automatically divide and merge contours without manual intervention.

Conclusion:

Level set methods provide more flexible and accurate segmentation than traditional snakes, especially when organ shapes change, split, or merge.